தீர்வு

(i) வரையறைப்படி $F(x) = P(X \le x) = \int_{-\infty}^{x} f(u)\,du$

$x < 1$ எனும்போது, $F(x) = P(X \le x) = \int_{-\infty}^{x} 0\,du = 0$.

$1 \le x < 2$ எனும்போது, $F(x) = P(X \le x) = \int_{-\infty}^{1} 0\,du + \int_{1}^{x} (u-1)\,du$

$$= 0 + \left[\frac{(u-1)^2}{2}\right]_1^x = \frac{(x-1)^2}{2}$$

$2 \le x < 3$ எனும்போது, $F(x) = P(X \le x) = \int_{-\infty}^{1} 0\,du + \int_{1}^{2} (u-1)\,du + \int_{2}^{x} (3-u)\,du$

$$= 0 + \left[\frac{(u-1)^2}{2}\right]_1^2 + \left[-\frac{(3-u)^2}{2}\right]_2^x$$

$$= \frac{1^2 - 0}{2} + \frac{1-(3-x)^2}{2} = 1 - \frac{(3-x)^2}{2}$$

$x \ge 3$ எனும்போது, $F(x) = P(X \le x) = \int_{-\infty}^{1} 0\,du + \int_{1}^{2} (u-1)\,du + \int_{2}^{3} (3-u)\,du + \int_{3}^{x} 0\,du$

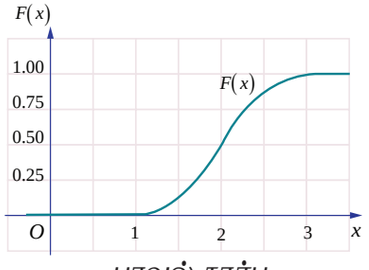
$$= \int_{-\infty}^{1} 0\,du + \int_{1}^{2} (u-1)\,du + \int_{2}^{3} (3-u)\,du + \int_{3}^{x} 0\,du$$

$$= 0 + \left[\frac{(u-1)^2}{2}\right]_1^2 + \left[-\frac{(3-u)^2}{2}\right]_2^3 + 0$$

$$= \frac{1}{2} + \frac{1}{2} = 1.$$

இவற்றின் மூலம்

$$F(x) = \begin{cases} 0, & -\infty < x < 1 \\ \dfrac{(x-1)^2}{2}, & 1 \le x < 2 \\ 1 - \dfrac{(3-x)^2}{2}, & 2 \le x < 3 \\ 1 & 3 \le x < \infty \end{cases}$$



பரவல் சார்பு
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(ii) $P(1.5 \le X \le 2.5) = F(2.5) - F(1.5)$

$$= \left(1 - \frac{(3-2.5)^2}{2}\right) - \left(\frac{(1.5-1)^2}{2}\right)$$